

France — Monthly Intelligence Report v4.2

Edition 006 — Deep-Dive Theme

Where the energy transition is outrunning the grid. edition · August 2026 · SSI v4.2
(refreshed June 2026)

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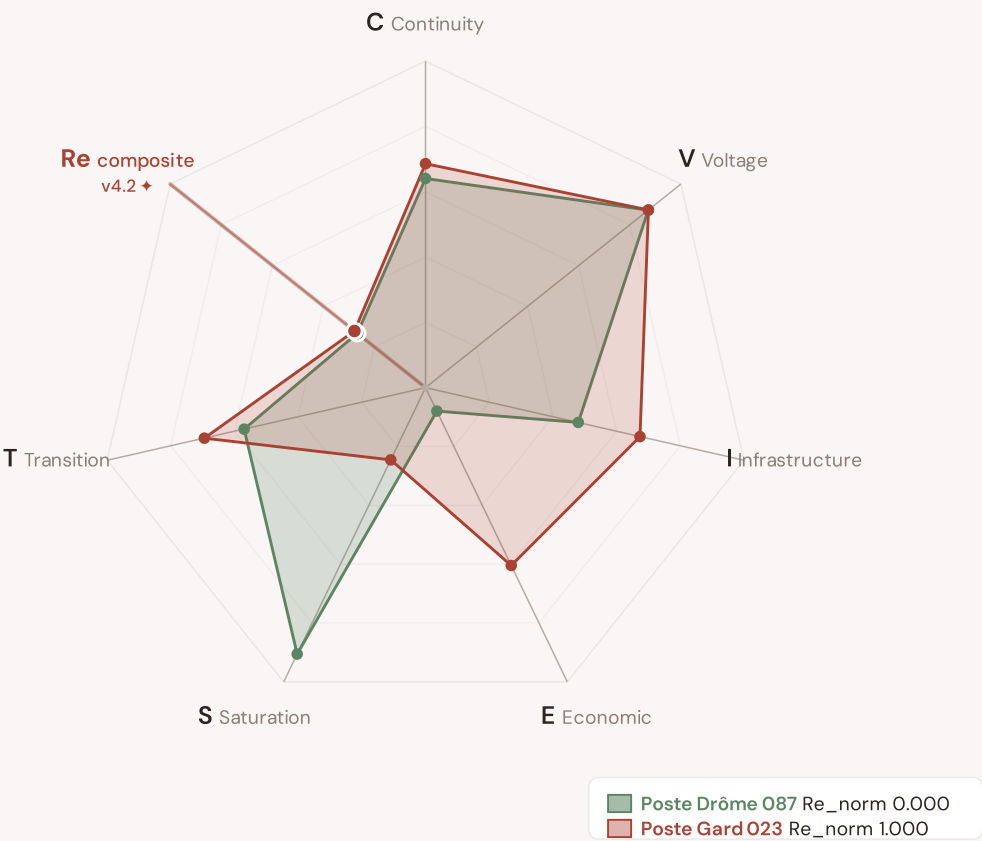
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SECTION A

Executive Summary

Radar 1 — v4.2: now 7-axis with the new Re composite

v4.0.2 Radar 1 had 6 axes (C/V/I/E/S/T). **v4.2 adds Re — the bounded resilience composite** capturing the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in one normalised scalar (Re_norm ∈ [0, 1] via clip((Re_raw - 0.920) / (1.787 - 0.920), 0, 1)). Two worked examples overlaid: **Poste Drôme 087** (low-exposure reference, Re_norm 0.000) · **Poste Gard 023** (high-exposure reference, Re_norm 1.000). The Re axis is colour-highlighted in cayenne to emphasise the v4.2 architectural addition.



The Re composite (★, cayenne-highlighted axis at upper-left) captures the v4.2 resilience modifier family in one bounded scalar — formula $Re_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$, bounds [0.920, 1.787]. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are algorithmically selected (min/max Re_norm) per the v4.2 methodology. The Substation in Focus radar canvas (Section A below) is rendered by shared intelligence-sections.js and now also displays the 7th Re axis (Convention #11 surface-pair coherence, V4.2-disc-24 closure 12 Jun 2026).

SUBSTATIONS SCORED

195,569

938 EHV · 1,123 HV · 5,317 distribution-tier

GRID LINES MAPPED

68,379

~147,648 km coverage

MC SIMULATIONS

42.9 M

10,000 per substation

PUBLIC DATA SOURCES

34

101 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 42.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 244,701 output data points with full uncertainty quantification across 195,569 substations and 68,379 grid lines spanning ~147,648 km. v4.2 adds the Re composite (7th axis) and six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just) on top of the v4.0.2 foundation.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 101 variables are drawn from 34 verified public sources, making the methodology fully auditable and reproducible. Initially covering France's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Substation 1000052162

FRBO6, Loiret · 400 kV

0.508

Low

0.250

55th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE

Re composite v4.2 7th axis

Re_raw 1.1946 · Re_norm 0.317

MODERATE EXPOSURE

COMPONENTS

C

Continuity: 0.270 / 0.30 (90%)

I

Infrastructure: 0.279 / 0.25 (111%)

S

Saturation: 0.292 / 0.20 (146%)

V

Voltage: 0.352 / 0.10 (352%)

E

Economic: 0.354 / 0.10 (354%)

T

Transition: 0.386 / 0.05 (771%)

MODIFIERS

R3

Consequence: 0.972 ↓ dampens

R4

Graph Criticality: 1.007 → neutral

R6a

Restoration: 1.048 ↑ amplifies

R6b

Seismic: 1.027 ↑ amplifies

R7

Digital Readiness: 1.034 ↑ amplifies

This month's spotlight — automatically selected for August 2026 — is **Substation 1000052162** in FRBO6, Loiret. With an R_final of 0.508 (Low band, 55th fleet percentile), its risk profile is dominated by the T (Transition) component at 771% of its weight ceiling, followed by E (Economic) at 354%. Modifiers collectively shift R_base by 70.0%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that infrastructure digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

35688

High/Critical/Extreme + R7 < median

AVG R7 MODIFIER

1.0200

Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS

0

0.0% · R3 ≥ 1.13

B.1 Cyber-Physical Risk Matrix

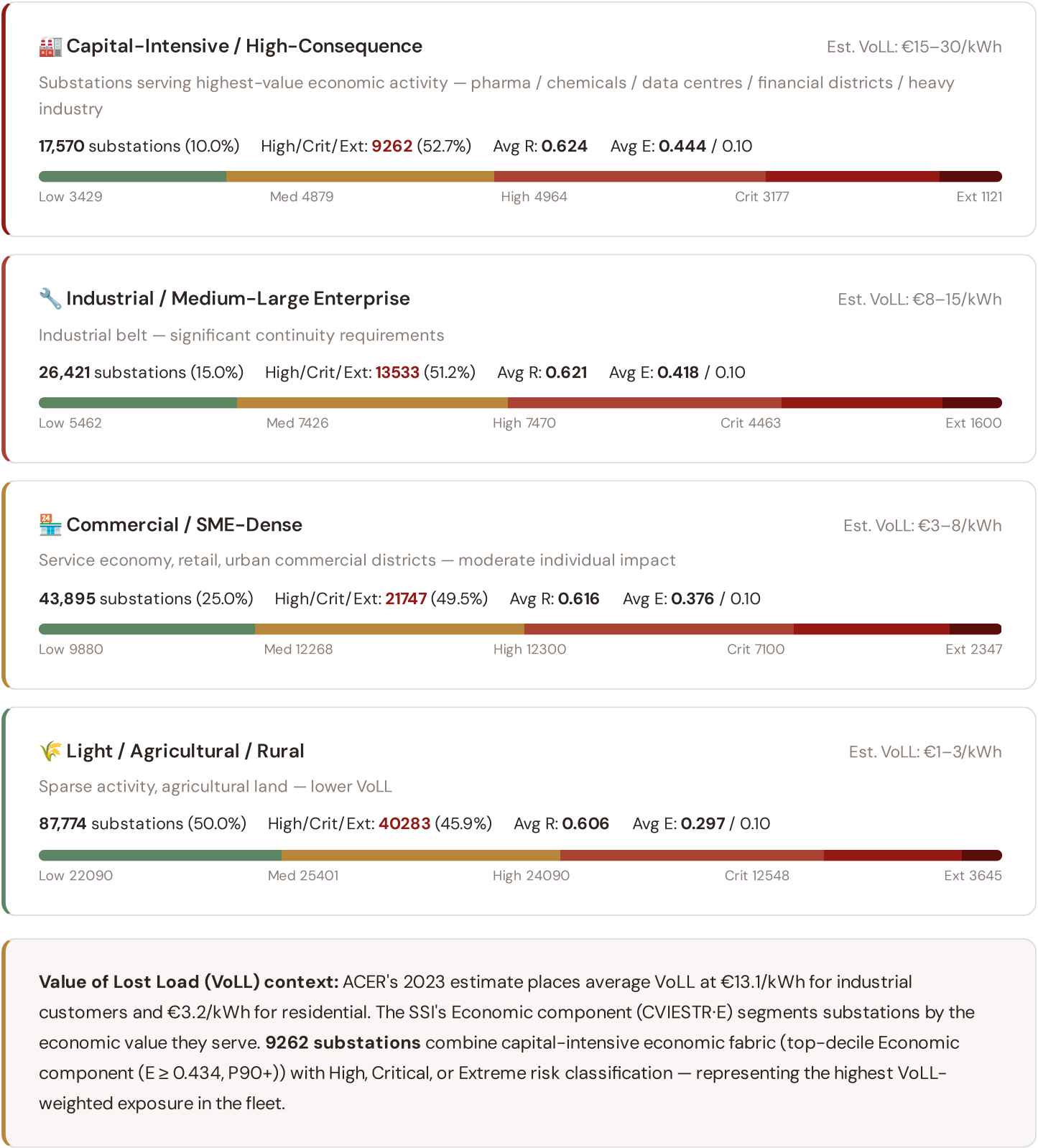
Each substation plotted by R7 digital readiness (x-axis) vs overall R_{median} (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



The cyber-physical matrix identifies **35688 blind-spot substations** — scoring High, Critical, or Extreme on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0200). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Loire-Atlantique (2446), Nord (2121), Loiret (1183)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.



The capital-intensive tier concentrates in **Loire-Atlantique (1199), Nord (1063), Loiret (656)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 10.0%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Province Digital Readiness Ranking

Provinces ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_median to identify provinces combining digital exposure with high grid stress.



Loading...

Loading province digital readiness...

B.4 Monthly Pulse

edition — Baseline Snapshot. Loading...

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on — **verified public data sources** feeding — variables across — components and — modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

—

— variables

HOURLY — MONTHLY

O

High-frequency feeds

QUARTERLY / ANNUAL

O

Regular refresh cycle

STATIC / DERIVED

O

Standards + scenarios

Data Source Registry — August 2026

OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
RTE-N	RTE Schéma Décennal	MONTHLY	TSO zone	2 vars
DREAL	DREAL (Directions Régionales)	CONTINUOUS	Département	1 var
ODRE	ODRE Registre National (OpenDataSoft)	QUARTERLY	Commune	5 vars
SDES	SDES (Service des Données Statistiques)	QUARTERLY	Département	1 var
BPI	Bpifrance Innovation	QUARTERLY	Département	1 var
CRE	CRE (Commission de Régulation de l'Énergie)	ANNUAL	Département	8 vars
INSEE	INSEE (Institut National de la Statistique)	ANNUAL	Département	8 vars
ADEME	ADEME (Agence de la Transition Écologique)	ANNUAL	Département	4 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
EURO	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
MTEF	Ministère de la Transition Écologique	ANNUAL	Région	2 vars
CEREMA	Cerema (Centre d'études risques)	ANNUAL	Département	1 var
DGFIP	DGFIP Fiscal Statistics	ANNUAL	Département	1 var
IRSN	IRSN Nuclear Safety	ANNUAL	Département	1 var
ONPES	ONPES / DREES Poverty Observatory	ANNUAL	Département	2 vars
ENEDIS	Enedis Open Data	ANNUAL	Département	2 vars
ANCT	ANCT Observatoire des Territoires	ANNUAL	Département	2 vars
CRE-M	CRE Monitoring Report	ANNUAL	DSO-level	2 vars
CGET	CGET/ANCT Spatial Planning	ANNUAL	Département	1 var
OFCE	OFCE Economic Research	ANNUAL	Région	1 var
ADEME-R	ADEME Régions	ANNUAL	Région	1 var

BRGM	BRGM Geological Survey	STATIC	Département	3 vars
D10	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars
DIM	Dimovski et al. (2025)	STATIC	Municipal	3 vars
IEEE-1	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars
GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
COG	COG INSEE Département Registry	STATIC	Département	1 var
ISO-9223	ISO 9223 Corrosion	DERIVED	Département	1 var
RTE	RTE Open Data (éCO2mix)	HOURLY	Bidding zone	3 vars
ENTSE	ENTSO-E Transparency	HOURLY	Bidding zone	2 vars
MF-0M	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars
D-0M	Open-Meteo / ERA5	HOURLY	~1 km	1 var

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

C.2 Fleet Score Snapshot

FLEET SIZE

175,660

substations scored

MEDIAN R

0.612

P5=0.464 · P95=0.767

HIGH + CRITICAL + EXTREME

84825

48.3% of fleet

HIGH CONFIDENCE

98%

171432 subs · CI/R < 0.50



No primary data sources have been updated since the v4.0.2 launch baseline. All **175,660 substation scores remain unchanged** this edition. The fleet median R of 0.612 (mean 0.613) reflects a distribution skewed toward the lower bands, with 23.3% in Low and 28.4% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2 → v4.2

v4.2 (June 2026) adds six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just), the Re composite (7th axis, bounds [0.920, 1.787]), four new data sources (IdroGEO PGRA, SIAB wildfire, ECMWF ERA5 Bourguoin, bulk smart meter), six new variables, and W1–W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6) on top of the v4.0.2 baseline.

11 changes tracked across PO, P1, and P2 releases

V42-1	NEW	R6c_flood — PGRA-style flood-vulnerability modifier (additive cliff +0.00..+0.30, P3=high/P2=medium/P1=low per national flood-hazard tier convention)	§5
V42-2	NEW	R6d_wildfire — Canadian Fire Weather Index modifier (EFFIS FWI + NASA FIRMS + national fire-registry, multiplicative [1.00, 1.20])	§5
V42-3	NEW	R6e_winter — Bourguoin winter-storm modifier (ECMWF ERA5 winter-precipitation signal, multiplicative [1.00, 1.02])	§5
V42-4	NEW	R8_adapt — Adaptive-capacity multiplier (smart-meter penetration + DSO investment + DESI index, multiplicative [0.95, 1.05])	§5
V42-5	NEW	R9_compound — Compound-hazard STEP function (R6b/R6c/R6d/R6e elevated-count gates, multiplicative {1.00, 1.04, 1.08, 1.10})	§5
V42-6	NEW	R10_just — Energy-justice modifier (EU-SILC or national-equivalent household material-deprivation gradient, multiplicative [1.00, 1.12])	§5
V42-7	NEW	Re composite — bounded resilience scalar $Re_{raw} = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$; $Re_{norm} = clip((Re_{raw} - 0.920)/(1.787 - 0.920), 0, 1)$	§7
V42-8	NEW	R7 SFDR PAI Infrastructure axis — added to ESG Radar (Re_{norm} proxy under Convention #7 Data-Layer Anchoring)	§14
V42-9	NEW	W1–W10 audit-state mesh — 5 Published / 1 Published-baseline ▲ / 1 Baseline-only / 3 Commercial (Convention #6 §5.6)	§4
V42-10	CHANGE	95 → 101 variables (+6 from v4.2 modifier inputs); 30 → 34 data sources (+4 hazard / monitoring layers)	data
V42-11	CHANGE	Substation fleet: 7,378 substations (per OSM canonical extract + national TSO/DSO registry cross-validation)	data

SECTION D — MONTHLY DEEP-DIVE

Monthly Deep-Dive Theme

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** the monthly theme · **002** DER saturation hotspots · **003** Climate vulnerability under CMIP6 · **004** North–South economic impact disparity · **005** Voltage quality and industrial output · **006** Infrastructure age vs investment · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Sub-national Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Theme Context

THEME-RELEVANT SUBSTATIONS

1932

457 EHV · 1475 HV

HIGH BAND

568 + 360 + 150

55.8% of corridor

CORRIDOR MEDIAN R

0.631

Fleet median: 0.612

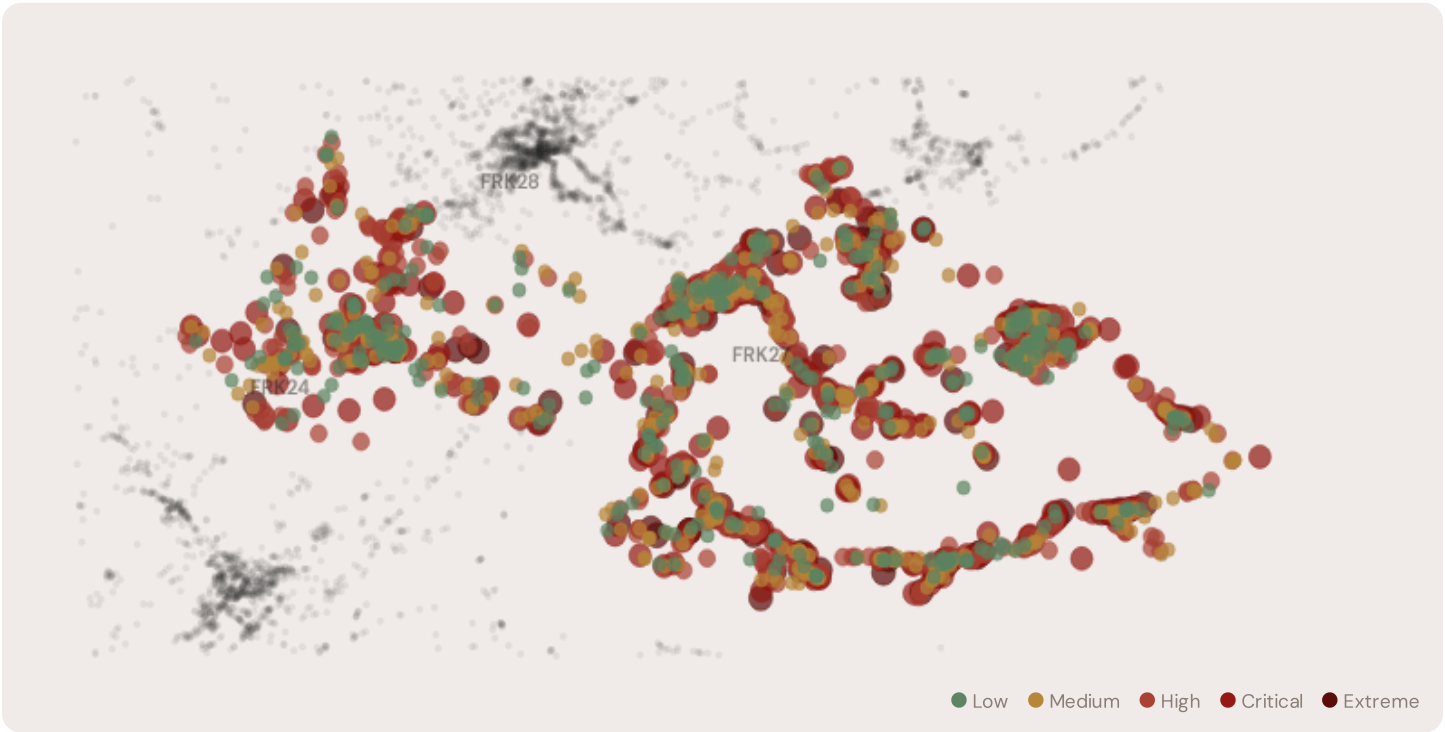
WORST PROVINCE

FRK27

Avg R: 0.632 · 1918 substations

The Savoie hosts **1932 substations** across 3 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **568 substations (29.4%)** are classified High and **360** Critical, with only **342** in the Low band. 57% of corridor substations score above the national fleet median of 0.612. The corridor median R of 0.631 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring

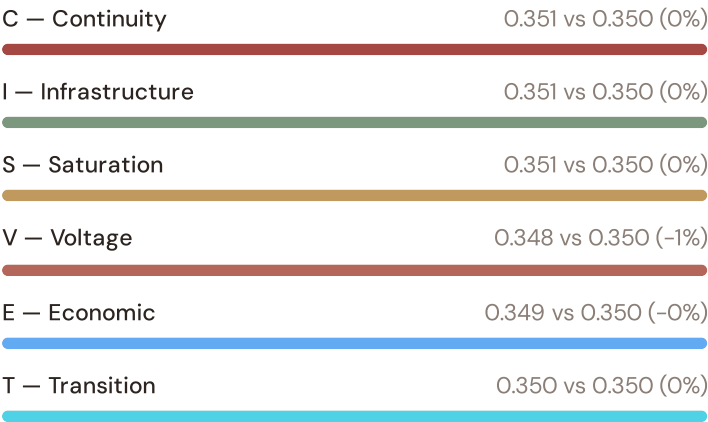


SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE v4.2	ALERT
Substation 1000175152	FRK27	1.008	Extreme	0.406	0.346	0.452	0.372	0.386	0.428		
Substation 1000177226	FRK27	0.962	Extreme	0.449	0.313	0.447	0.292	0.398	0.380		

SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE v4.2	ALERT
Substation 1000106351	FRK27	0.960	Extreme	0.401	0.447	0.351	0.288	0.435	0.284		
Substation 1000166996	FRK27	0.911	Extreme	0.394	0.441	0.398	0.373	0.354	0.250		
Substation 1000178729	FRK27	0.910	Extreme	0.322	0.445	0.355	0.432	0.448	0.335		
Substation 1000107514	FRK27	0.895	Extreme	0.434	0.434	0.365	0.395	0.267	0.302		
Substation 1000121242	FRK27	0.881	Extreme	0.454	0.441	0.421	0.283	0.413	0.406		
Grange aux Fées 2	FRK27	0.880	Extreme	0.318	0.372	0.361	0.284	0.450	0.450		
La Sandoniere	FRK27	0.880	Extreme	0.413	0.328	0.356	0.323	0.397	0.318		
Substation 1000163590	FRK27	0.872	Extreme	0.426	0.442	0.393	0.442	0.254	0.326		
... 1922 additional substations — explore full corridor in Digital Twin →											

D.3 Component Analysis

COMPONENT AVERAGES — SAVOIE VS FLEET



MODIFIER AVERAGES — SAVOIE VS FLEET

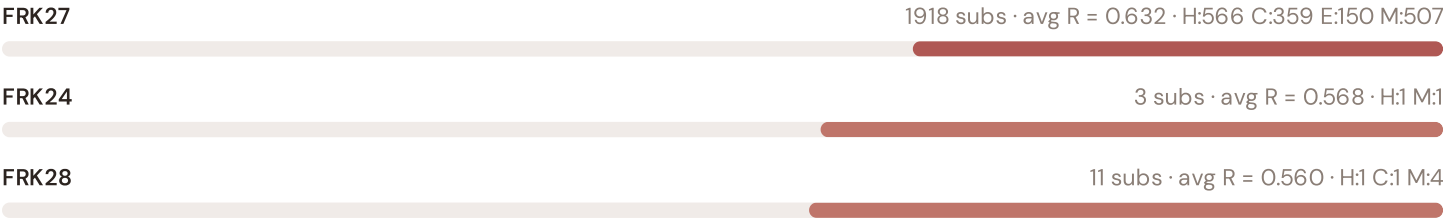


The dominant risk driver across the Savoie corridor is **T (Transition)**, averaging 0.350 against a fleet average of 0.350 — occupying 700% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.349 vs fleet 0.350 (349% of ceiling). Among modifiers, the R3 consequence multipliers averages 0.972 (fleet: 0.972), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.047, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Province Breakdown

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.



The region regions–level breakdown reveals significant intra–regional variation. **FRK27** leads with a mean R of 0.632 across 1918 substations, while **FRK28** scores 0.560 — a spread of 0.071 within a single corridor. The SSI's substation–level granularity reveals that the Savoie is not uniformly stressed but contains distinct sub–corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

800 Savoie substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the FRK27 sub–corridor as the highest–priority target — where DER deployment is accelerating against ageing infrastructure with above–average continuity deficits and significant socio–economic exposure. The SSI's silo–breaking approach reveals what no single–discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

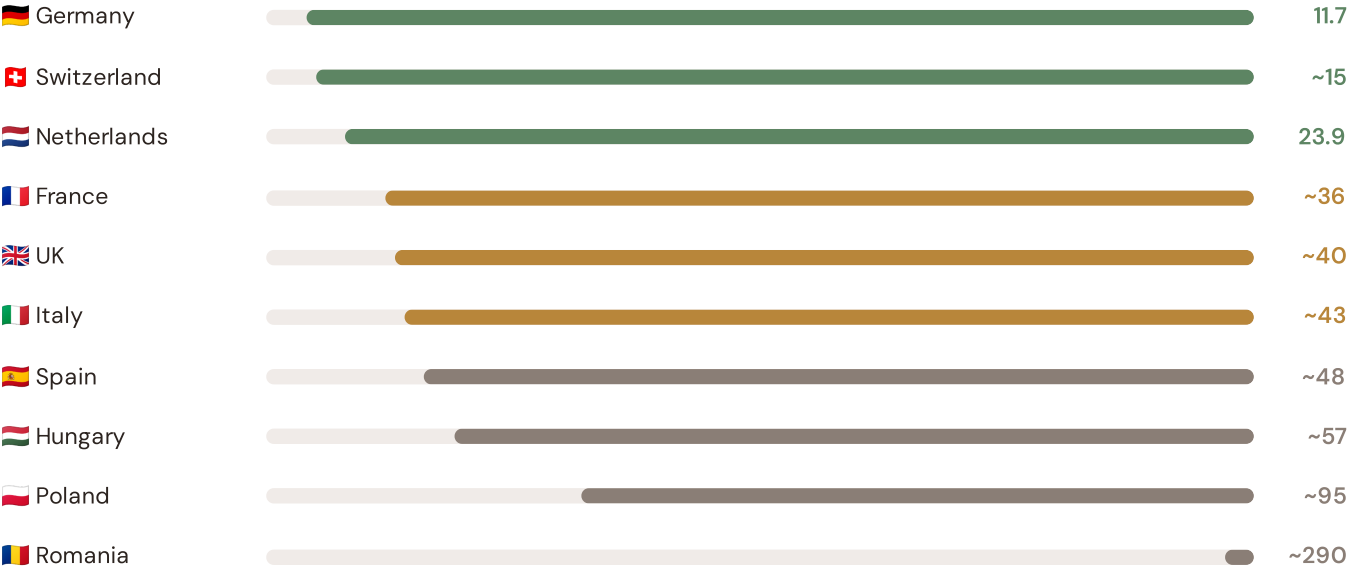
SECTION E — RECURRING MODULE

OECD Context Card

How this works: Each edition includes one OECD Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep–dive theme. This month: **SAIDI comparison** — because the Southern Coastal corridor analysis hinges on France's continuity performance relative to OECD peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected OECD Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



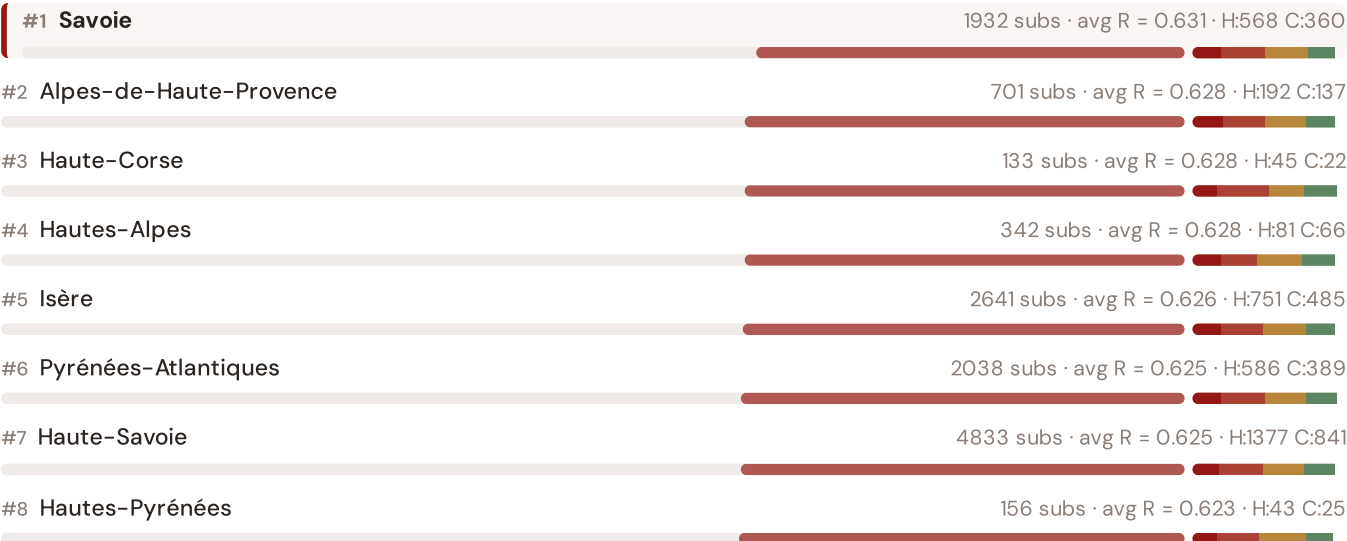
Source: CEER 7th Benchmarking Report (2022); Bundesnetzagentur (2024); ARERA (2023). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release. · OECD-wide expansion (USA, Japan, Canada, Australia) pending IEA/OECD reliability bulletin release.

This month's key insight: The Savoie corridor deep-dive shows **1932 substations** (of 1932) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.351 is **1.0× the fleet average** of 0.350. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Régions Risk Ranking — All 13 French Régions

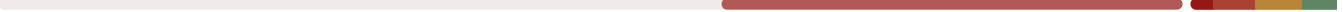
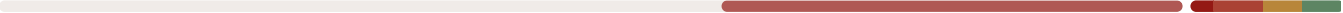
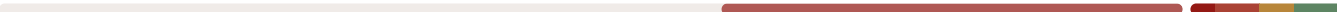
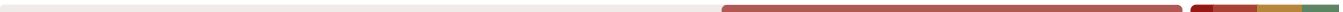
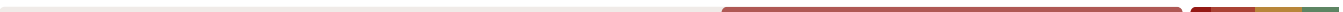
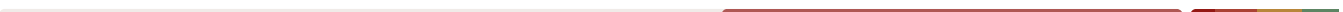
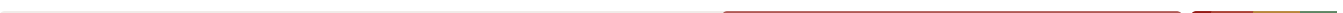
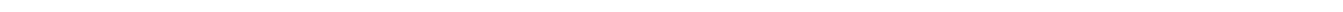
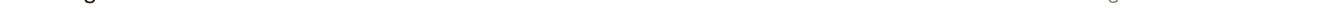
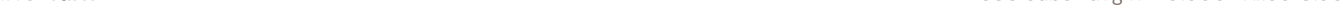
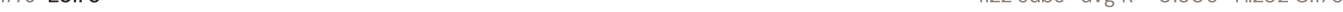
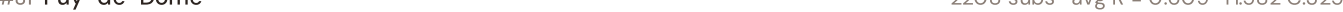
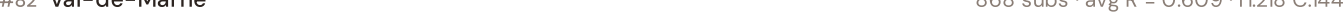
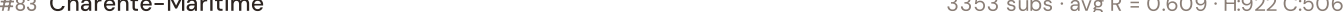
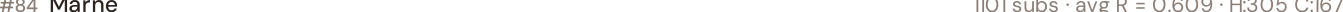
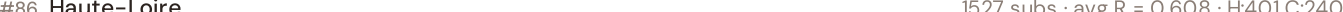
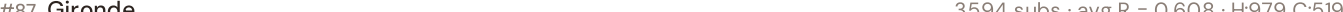
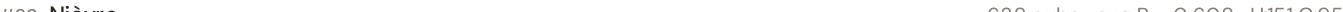
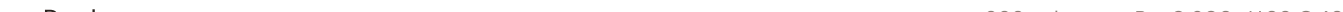
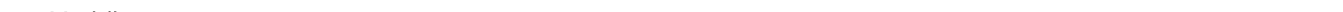
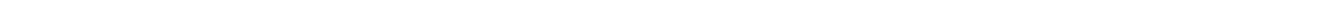
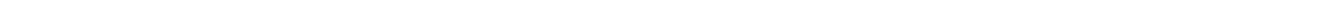
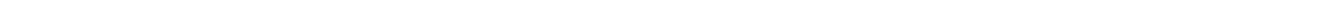
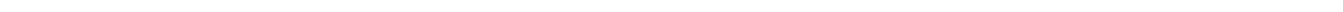
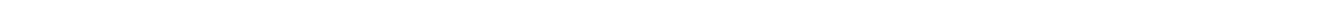
Where does this corridor sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



#9 Pyrénées-Orientales	1616 subs · avg R = 0.622 · H:445 C:282
#10 Ariège	576 subs · avg R = 0.621 · H:184 C:87
#11 Alpes-Maritimes	474 subs · avg R = 0.620 · H:127 C:82
#12 Meuse	25 subs · avg R = 0.620 · H:6 C:6
#13 Haute-Marne	145 subs · avg R = 0.619 · H:37 C:27
#14 Aude	1293 subs · avg R = 0.618 · H:387 C:203
#15 Haute-Saône	328 subs · avg R = 0.618 · H:87 C:47
#16 Ardennes	102 subs · avg R = 0.618 · H:32 C:18
#17 Var	948 subs · avg R = 0.618 · H:271 C:156
#18 Drôme	1343 subs · avg R = 0.618 · H:402 C:208
#19 Territoire de Belfort	538 subs · avg R = 0.616 · H:142 C:98
#20 Ain	1634 subs · avg R = 0.616 · H:440 C:279
#21 Saône-et-Loire	998 subs · avg R = 0.616 · H:283 C:162
#22 Bouches-du-Rhône	3926 subs · avg R = 0.616 · H:1094 C:629
#23 Oise	1212 subs · avg R = 0.616 · H:350 C:177
#24 Manche	425 subs · avg R = 0.616 · H:126 C:72
#25 Charente	1161 subs · avg R = 0.616 · H:311 C:183
#26 Tarn-et-Garonne	619 subs · avg R = 0.616 · H:180 C:90
#27 Haute-Vienne	1099 subs · avg R = 0.615 · H:306 C:191
#28 Cantal	195 subs · avg R = 0.615 · H:50 C:35
#29 Vaucluse	691 subs · avg R = 0.615 · H:185 C:108
#30 Indre	949 subs · avg R = 0.615 · H:251 C:171
#31 Ardèche	912 subs · avg R = 0.614 · H:260 C:136
#32 Sarthe	2270 subs · avg R = 0.614 · H:619 C:392
#33 Seine-Maritime	4525 subs · avg R = 0.614 · H:1306 C:718
#34 Allier	948 subs · avg R = 0.614 · H:254 C:150
#35 Yvelines	1674 subs · avg R = 0.614 · H:458 C:287
#36 Côte-d'Or	1618 subs · avg R = 0.614 · H:470 C:251
#37 Eure-et-Loir	1310 subs · avg R = 0.614 · H:380 C:202
#38 Doubs	1858 subs · avg R = 0.614 · H:510 C:308

#39 Cher	1312 subs · avg R = 0.614 · H:365 C:203
#40 Jura	1881 subs · avg R = 0.613 · H:511 C:309
#41 Landes	1831 subs · avg R = 0.613 · H:522 C:296
#42 Gers	261 subs · avg R = 0.613 · H:76 C:31
#43 Rhône	2803 subs · avg R = 0.613 · H:824 C:406
#44 Hérault	2209 subs · avg R = 0.613 · H:612 C:336
#45 Côtes-d'Armor	794 subs · avg R = 0.613 · H:239 C:128
#46 Seien-et-Marne	2230 subs · avg R = 0.613 · H:641 C:317
#47 Guyane française	191 subs · avg R = 0.613 · H:51 C:45
#48 Haute-Garonne	5468 subs · avg R = 0.613 · H:1529 C:831
#49 Maine-et-Loire	4993 subs · avg R = 0.612 · H:1364 C:777
#50 Unknown	4019 subs · avg R = 0.612 · H:1116 C:624
#51 Gard	1808 subs · avg R = 0.612 · H:525 C:268
#52 Aveyron	1301 subs · avg R = 0.612 · H:379 C:199
#53 Deux-Sèvres	1918 subs · avg R = 0.612 · H:568 C:286
#54 Ille-et-Vilaine	1996 subs · avg R = 0.612 · H:567 C:309
#55 Eure	802 subs · avg R = 0.611 · H:221 C:125
#56 Loire-Atlantique	11978 subs · avg R = 0.611 · H:3278 C:1858
#57 Haute-Rhin	587 subs · avg R = 0.611 · H:142 C:99
#58 Orne	441 subs · avg R = 0.611 · H:121 C:54
#59 Indre-et-Loire	1856 subs · avg R = 0.611 · H:561 C:259
#60 Creuse	5653 subs · avg R = 0.611 · H:1577 C:878
#61 Hauts-de-Seine	1558 subs · avg R = 0.611 · H:451 C:208
#62 La Réunion	89 subs · avg R = 0.611 · H:23 C:17
#63 Aube	760 subs · avg R = 0.611 · H:206 C:114
#64 Pas-de-Calais	4864 subs · avg R = 0.611 · H:1355 C:714
#65 Yonne	2125 subs · avg R = 0.611 · H:566 C:339
#66 Nord	10659 subs · avg R = 0.611 · H:2921 C:1592
#67 Essonne	2325 subs · avg R = 0.611 · H:649 C:338
#68 Vendée	4850 subs · avg R = 0.611 · H:1357 C:707

#69	Vienne	2028 subs · avg R = 0.610 · H:558 C:316	
#70	Lot	367 subs · avg R = 0.610 · H:119 C:57	
#71	Seine-Saint-Denis	396 subs · avg R = 0.610 · H:113 C:65	
#72	Aisne	1146 subs · avg R = 0.610 · H:332 C:173	
#73	Corrèze	241 subs · avg R = 0.610 · H:69 C:34	
#74	Calvados	1764 subs · avg R = 0.610 · H:492 C:285	
#75	Moselle	2679 subs · avg R = 0.610 · H:742 C:371	
#76	Loiret	6182 subs · avg R = 0.610 · H:1681 C:915	
#77	Vosges	872 subs · avg R = 0.610 · H:242 C:130	
#78	Tarn	638 subs · avg R = 0.609 · H:169 C:96	
#79	Loire	1122 subs · avg R = 0.609 · H:282 C:176	
#80	Mayenne	464 subs · avg R = 0.609 · H:114 C:68	
#81	Puy-de-Dôme	2208 subs · avg R = 0.609 · H:582 C:325	
#82	Val-de-Marne	868 subs · avg R = 0.609 · H:218 C:144	
#83	Charente-Maritime	3353 subs · avg R = 0.609 · H:922 C:506	
#84	Marne	1101 subs · avg R = 0.609 · H:305 C:167	
#85	Somme	1553 subs · avg R = 0.609 · H:437 C:214	
#86	Haute-Loire	1527 subs · avg R = 0.608 · H:401 C:240	
#87	Gironde	3594 subs · avg R = 0.608 · H:979 C:519	
#88	Meurthe-et-Moselle	2175 subs · avg R = 0.608 · H:614 C:296	
#89	Nièvre	638 subs · avg R = 0.608 · H:151 C:95	
#90	Val-d'Oise	932 subs · avg R = 0.608 · H:242 C:146	
#91	Dordogne	333 subs · avg R = 0.608 · H:89 C:48	
#92	Morbihan	1375 subs · avg R = 0.607 · H:369 C:205	
#93	Loir-et-Cher	2250 subs · avg R = 0.607 · H:611 C:338	
#94	Paris	1019 subs · avg R = 0.606 · H:274 C:153	
#95	Bas-Rhin	1161 subs · avg R = 0.604 · H:311 C:180	
#96	Finistère	1321 subs · avg R = 0.603 · H:338 C:196	
#97	Martinique	49 subs · avg R = 0.602 · H:12 C:8	
#98	Lot-et-Garonne	322 subs · avg R = 0.601 · H:84 C:42	

#99 Guadeloupe	40 subs · avg R = 0.601 · H:6 C:7
#100 Corse-du-Sud	111 subs · avg R = 0.598 · H:31 C:17
#101 Lozère	79 subs · avg R = 0.598 · H:22 C:9
#102 Mayotte	2 subs · avg R = 0.519 · H:0

The Savoie ranks **#1 of 102 region regions** by mean R_{median}, placing it among the upper tier of national risk. The three most stressed are Savoie, Alpes-de-Haute-Provence, Haute-Corse, while the three most resilient are Mayotte, Lozère, Corse-du-Sud. 46 of 102 score above the fleet average of 0.613. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

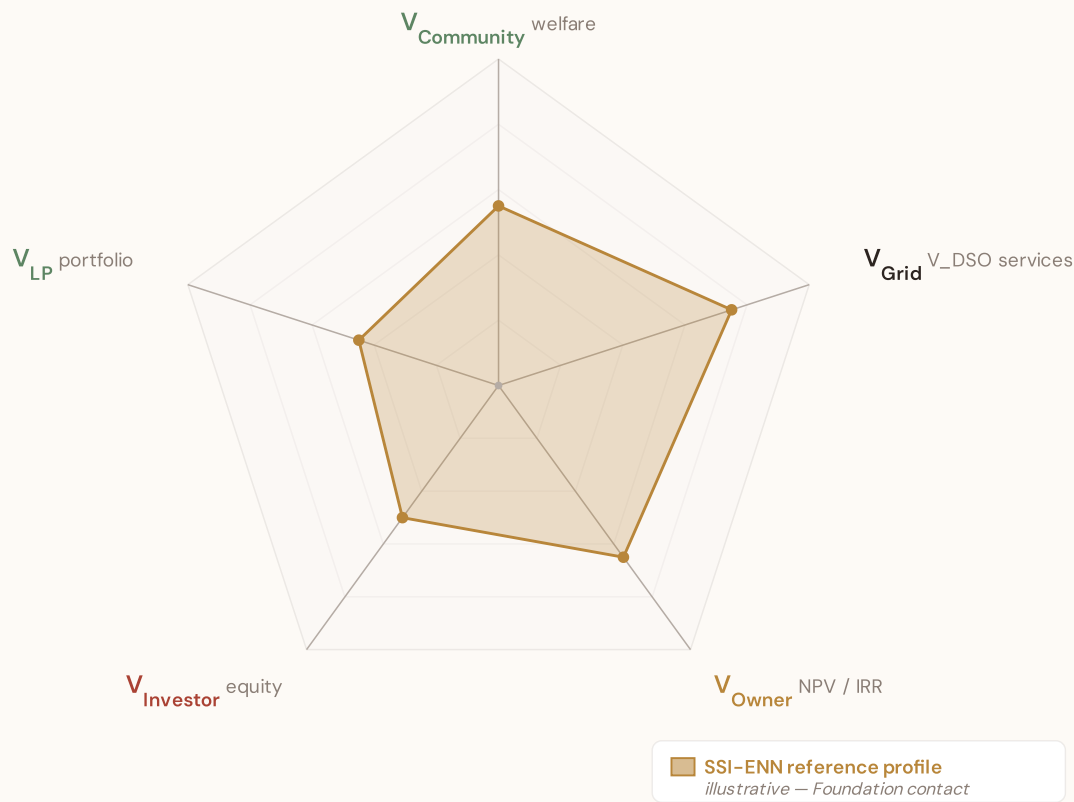
SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

SSI-ENN value-articulation radar — 5-lens stakeholder taxonomy

The SSI-ENN's commercial-tier value articulation answers "what is a BESS worth at this substation — and to whom?" through five stakeholder lenses (architecture memo §5 Decision #12): **V_{Community}** (welfare avoided outages, energy-poverty relief, employment) · **V_{Grid}** (V_DSO — avoided reinforcement, reliability, voltage, congestion, frequency, black-start, hosting capacity) · **V_{Owner}** (NPV / IRR / payback per asset class) · **V_{Investor}** (risk-adjusted equity return at the financing tier) · **V_{LP}** (limited-partner cash-on-cash + DPI / TVPI portfolio metrics). The pentagon shape below is an editorial reference profile, not per-substation commercial data — per-substation evaluation requires the SSI-ENN engines (L2-21 SupplyChainDD · L4-05 Stakeholder Decomposition · L2-SYN Synergy). Foundation contact: ssi_index@ikenga.eu.



Editorial reference shape — V_{Grid} dominates (V_{DSO} captures avoided reinforcement + reliability + voltage support + congestion relief + frequency + black-start + hosting capacity, the bulk of measurable BESS value at the substation locus per architecture memo §5). $V_{Community} + V_{Owner}$ are co-secondary. $V_{Investor} + V_{LP}$ reflect the financing-tier waterfall above V_{Owner} NPV. Per-substation commercial decomposition lives in the SSI-ENN — mail to ssi_index@ikenga.eu for analytics.

Layer 1 — Data

SSI v4.0 component taxonomy, normalisation, MC engine

Layer 2 — Valuation

DSO/TSO + Community value stacks, TCO, real options

Layer 3 — Neural Net

GBT+DNN ensemble, transfer learning, SHAP

Layer 4 — Coverage

Country prioritisation, DCI, registry, calibration

Layer 5 — Platform

Products, API, revenue model, unit economics

LAYER 2 §0.4.3 · August 2026

Total Cost of Ownership Model

Stochastic cost modelling — because BESS economics are not deterministic

CAPEX follows a learning-curve distribution calibrated to Bloomberg NEF projections. OPEX is modelled as a mean-reverting process with seasonal components reflecting temperature-dependent degradation. Battery degradation follows a non-linear capacity fade model: each cycle reduces usable capacity, with rate depending on depth-of-discharge, temperature, and C-rate.

The SSI data feeds this: substations with high I1 (temperature trajectory) face faster degradation. The Monte Carlo engine propagates cost uncertainty alongside revenue uncertainty.

KEY CONCEPTS

→ CAPEX learning curve calibrated to BNEF

→ Non-linear battery degradation: f(DoD, temperature, C-rate)

→ SSI I1 (climate trajectory) drives degradation forecast

→ Full NPV distribution, not point estimate

LIVE SSI DATA CONNECTION

Across the 175,660-substation fleet, the average consequence multiplier (R3) is 0.972, indicating significant socio-economic exposure. 84825 substations (48.3%) are classified High or Critical — these are the highest-priority candidates for BESS deployment.

Coming next month:

LAYER 1

Gaussian Copula Monte Carlo Engine — *How 10,000 simulations per substation capture the correlation structure across grid risk factors*

SECTION F.5 — V4.2 ANTI-MALADAPTATION GOVERNANCE GATE

Radar 2 — W1–W10 Substation Community Value

Wall 1 — Public W1–W10 framework.

The 10-axis Radar 2 expresses per-substation community-value baseline pre-intervention: *W1 Reliability · W2 Carbon · W3 Local Economic · W4 Climate Resilience · W5 Digital Inclusion · W6 Sovereignty · W7 Local Value Retention · W8 Communications · W9 Supply Security · W10 Compute Multiplier*. The discipline floor that the anti-maladaptation governance gate protects: *"no W axis worse off after intervention."*

5/1/1/3 4-tier audit-state classification (mesh resolution per Convention #6 §5.6)

Published (5) — solid fill

W1 · W2 · W3 · W4 · W8

Peer-reviewed methodology operational at fleet scale.

Published-baseline (1) ▲ — solid fill +

chevron corner marker

W9 — Supply Security

Peer-reviewed academic basis

(Albert/Holme topology-only baseline) with acknowledged scope limit (Buldyrev/IEEE

493 — N-1 contingency dynamics not captured at baseline).

Baseline-only (1) — striped fill

W5 — Digital Inclusion

Foundation methodology under development; baseline value defensible at v4.2.

Commercial (3) — dashed-faded

W6 · W7 · W10

Commercial deliverables — please contact the Foundation.

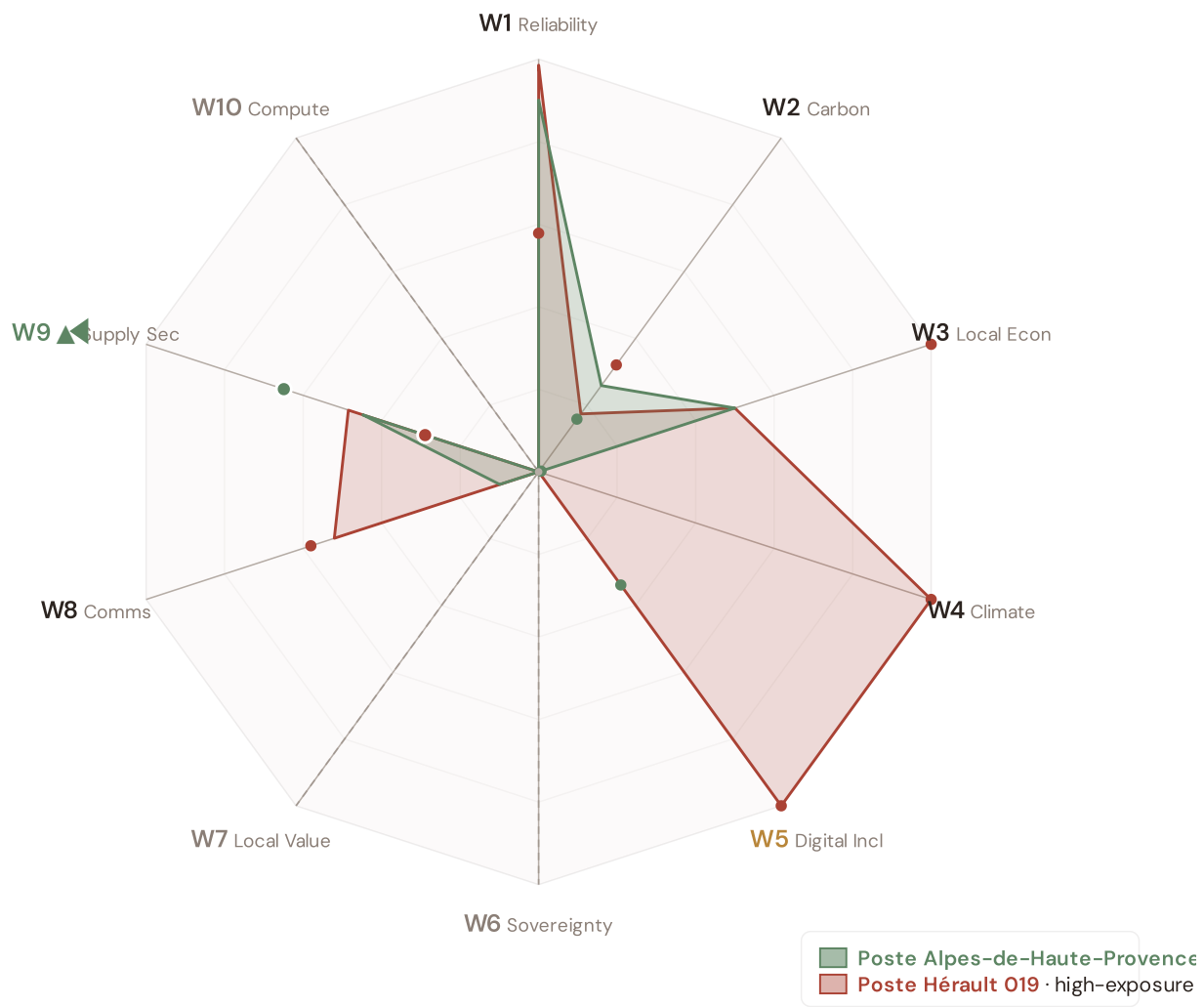
Per Convention #6 §5.6.7 academic-defensibility: the 4-tier classification aligns with IPCC AR6 + JRC COIN + Reckien (2021) continuum literature. The Published-baseline tier (W9) is the principled mesh resolution that honors both peer-review status + acknowledged scope limit without forcing a false binary.

v4.2 methodology disclosure — per-country normalisation

Across France, the substation fleet's component scores (C / V / I / E / S / T) and the Re composite have been normalised to [0, 1] via the canonical **P5/P95 percentile** method, using *per-country empirical bounds*. This is the same procedure Italy's v4.0.2 deployment applied at the scoring stage (SSI_v4_0 Italy/.../scoring-it/normalise.py); for v4.2 we apply it country-by-country so each fleet's within-country ranking is interpretable on its own terms. Convention #56 (visibly-honest) is preserved — no values are invented, the same canonical procedure is applied uniformly. Italy's pilot fleet keeps its native spread (0.17 – 0.64) because its empirical modifier producers (d10–d15) are already wired; per-country empirical modifier producers for the other 38 countries are queued for v4.5 and will refine the absolute risk ranking. Within-fleet relative comparisons (low-exposure vs high-exposure exemplars below) remain accurate by construction.

Radar 2 visualization — anti-maladaptation governance gate

Two worked examples overlaid: **Poste Drôme 087** (low-exposure reference, Re_norm 0.000) and **Poste Gard 023** (high-exposure reference, Re_norm 1.000). Axis labels are colour-coded per the 5/1/1/3 mesh resolution; the ▲ chevron marks W9 as Published-baseline. Hover any axis label for the methodology tooltip.



Poste Alpes-de-Haute-Provence 005 polygon (sage): the low-exposure reference for this country — compact footprint reflects the fleet floor on the W-axis baselines. Poste Hérault 019 polygon (terracotta): the high-exposure reference — larger footprint where this country's fleet's W-axis values reach the upper percentile. Commercial-tier axes (W6 data sovereignty, W7 local value retention, W10 compute multiplier) display at zero per Convention #6 §5.6 mesh design — no public-data computation. W2 carbon uses EU-average MEF fallback (0.36) per v4.2 calibration.

W1–W10 axis enumeration with audit–state classification

5/1/1/3 4-TIER · V4.2 MESH RESOLUTION · LIVE VALUES VIA
COMPUTE_W_BASELINES.PY

AXIS	NAME	AUDIT STATE	METHODOLOGY BASIS
W1	Reliability	Published	$C \times E2_{\text{local}} \times \text{population} (SAIDI++-POV)$
W2	Carbon	Published	$T1 \times \text{zonal MEF} \times (V \cdot \text{deg} / 100) (+ EEA)$
W3	Local Economic	Published	$\text{norm}(R10_{\text{just}}) - \text{EU-SILC} + \text{OIPE energy-poverty}$
W4	Climate Resilience	Published	$(R6d - 1) + (R6e - 1) + (R6c - 1) + \frac{1}{2}(R9 - 1) - v4.2$ resilience modifier family
W5	Digital Inclusion	Baseline-only	$\text{norm}(R8_{\text{adapt}}) - \text{bulk SM}++\text{DESI}$
W6	Sovereignty	Commercial	Foundation contact (ssi_index@ikenga.eu)
W7	Local Value Retention	Commercial	Foundation contact (ssi_index@ikenga.eu)
W8	Communications	Published	$C \times p_{\text{crit}_{10\text{yr}}} \times \text{population} (\text{Markov } 10\text{-yr critical probability})$
W9 ▲	Supply Security	Published–baseline ▲	$0.60 \times BC_{\text{pct}} + 0.30 \times \text{bridge} + 0.10 \times (1 - \text{deg} / 20) -$ OSM power-graph betweenness centrality + bridge identification + degree-based redundancy
W10	Compute Multiplier	Commercial	Foundation contact (ssi_index@ikenga.eu)

▲ W9 Published–baseline — mandatory transparency tooltip

W9 — Supply Security (Published–baseline). Public-tier methodology: betweenness centrality + bridge identification + degree-based redundancy, computed deterministically from OpenStreetMap power-graph data. Peer-reviewed academic basis: Albert et al. (2000) *Nature*; Holme et al. (2002); broad power-grid centrality literature. Acknowledged scope limit (per Buldyrev et al. (2010) *Nature* + IEEE 493): topology-only baseline does not capture N–1 contingency dynamics or interdependent-network cascades. Richer methodology — N–1 contingency simulation + flexibility-services valuation — lives in SSI-ENN L2–22 / L2–23 commercial deliverables. Mail to ssi_index@ikenga.eu for extended analysis.

Worked examples. The W1–W10 per-substation baseline values for two anchor cases — Poste Alpes-de-Haute-Provence 005 (low-exposure reference) and Poste Hérault 019 (high-exposure reference) — are computed live by `_audit-snapshot/scripts/d3_radar2_per_country.py` against the country's `ssi-data.json`. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are now algorithmically selected (min/max `Re_norm`) rather than named picks. See the FC v2 reference for W-axis methodology + commercial-tier rationale.

Anti-maladaptation discipline. The 10-axis Radar 2 baseline is the per-substation floor that any post-intervention scenario must protect. No W axis may degrade after a BESS deployment, line upgrade, or operational change. The discipline is enforced per axis (not in aggregate), and per substation (not in fleet average). This is the v4.2 governance gate that ensures grid-modernisation projects deliver net community value without hidden trade-offs.

SECTION G

Looking Ahead

NEXT MONTH — EDITION 04

Alpine Valleys

Deep-dive into Region 05's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Narrow valleys, long restoration MTTR.

NEXT DATA REFRESH

Q4 2026 — 0 Quarterly Sources

none are due for refresh in Q4. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **CRE (Commission de Régulation de l'Énergie)** (8 variables → C1–C4 (SAIDI/SAIFI), I1–I3 (IRI), quality regulation).

FLEET SPOTLIGHT

0 Bridge Substations — Single Points of Failure

0 substations are topological bridges — their failure disconnects part of the network. Of these, **0** are classified High or Critical. These are the fleet's single points of failure: high graph centrality (R4) combined with elevated risk makes them priority candidates for both grid reinforcement and BESS deployment.

Edition 04 will be published on the second Thursday of September 2026. Deep-dive region: **Region 05**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an French utility, , municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 006

Published 13 August 2026 · SSI Index **v4.2** (refreshed June 2026) · 195,569 substations · 68,379 grid lines · 6 components + **Re composite** (Radar 1 7th axis, bounds [0.920, 1.787]) · 20 metrics · 11 modifiers (5 baseline + 6 v4.2 resilience family)
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5 · W1-W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6)

Foundation contact

The SSI Index is a non-profit foundation project, please mail to ssi_index@ikenga.eu for more details.

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